Proving the Limit Laws

Examples. (Problems on next page)

(1) $\lim_{x\to a} cf(x) = c\lim_{x\to a} f(x)$

Proof. Let $L_1 := \lim_{x\to a} f(x)$. If $c = 0$, then $cf(x) = 0$ for every x in the domain of f, so $\lim_{x\to a} cf(x) = \lim_{x\to a} 0 = 0 = 0 \cdot \lim_{x\to a} f(x)$. This shows that equality holds if $c = 0$. Now suppose $c \neq 0$. Then, given $\varepsilon > 0$, corresponding to $\frac{\varepsilon}{|c|} > 0$ there exists $\delta > 0$ such that $0 < |x-a| < \delta$ implies $|f(x) - L_1| < \frac{\varepsilon}{|c|}$. It follows that if $0 < |x-a| < \delta$ we have

$$|cf(x) - cL_1| = |c(f(x) - L_1)| = |c||f(x) - L_1| < |c|\frac{\varepsilon}{|c|} = \varepsilon.$$

This proves that $\lim_{x\to a} cf(x) = cL_1 = c\lim_{x\to a} f(x)$. □

(2) $\lim_{x\to a}(f(x) + g(x)) = \lim_{x\to a} f(x) + \lim_{x\to a} g(x)$

The proof uses the *Triangle Inequality*, which states that $|x+y| \leq |x| + |y|$ for every $x, y \in \mathbb{R}$.

Proof. Let $L_1 := \lim_{x\to a} f(x)$ and $L_2 := \lim_{x\to a} g(x)$. Let $\varepsilon > 0$. Since $\lim_{x\to a} f(x) = L_1$ there exists $\delta_1 > 0$ such that $0 < |x-a| < \delta_1$ implies $|f(x) - L_1| < \frac{\varepsilon}{2}$. Since $\lim_{x\to a} g(x) = L_2$ there exists $\delta_2 > 0$ such that $0 < |x-a| < \delta_2$ implies $|f(x) - L_2| < \frac{\varepsilon}{2}$. Then $0 < |x-a| < \min\{\delta_1, \delta_2\}$ implies

$$\begin{aligned} |f(x) + g(x) - (L_1 + L_2)| &= |f(x) - L_1 + g(x) - L_2| \\ &\leq |f(x) - L_1| + |g(x) - L_2| \quad \text{(by the Triangle Inequality)} \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

This proves that $\lim_{x\to a}(f(x) + g(x)) = L_1 + L_2 = \lim_{x\to a} f(x) + \lim_{x\to a} g(x)$. □

1. Prove that $\lim_{x\to a} f(x) = L$ if and only if $\lim_{x\to a}(f(x) - L) = 0$.

2. Prove that $\lim_{x\to a} f(x)g(x) = \lim_{x\to a} f(x) \lim_{x\to a} g(x)$. *Hint*: You may use the identity

$$f(x)g(x) - L_1L_2 = (f(x) - L_1)(g(x) - L_2) + L_1(g(x) - L_2) + L_2(f(x) - L_1),$$

where $L_1 := \lim_{x\to a} f(x)$ and $L_2 := \lim_{x\to a} g(x)$. By Question 1, it is sufficient to show that $\lim_{x\to a}(f(x)g(x) - L_1L_2) = 0$.